A National Innovation Operating System: Converting Scientific and Human Capital into Economic Value in the Republic of Kazakhstan

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Abstract

Kazakhstan has set the strategic objective of diversifying its economy beyond extractive industries toward a knowledge-based economy driven by science, technology, and a modernized workforce. The binding constraint on that transition is not the absence of scientific or human capital, but the absence of an *instrument* that can measure it, identify where it is most valuable, and convert it into economic activity at national scale. This proposal delivers that instrument. Building on a validated machine-learning approach to identifying commercially valuable science (Masclans et al.), a deployed research-scoring platform, a workforce-redesign engine developed under an OpenAI research grant, and an authoring and delivery infrastructure for AI-based learning, we propose a *National Innovation Operating System* (NIOS): an integrated set of AI instruments that (i) score the commercial, scientific, and social potential of the nation's entire research base; (ii) assemble a fundable, diversified deep-technology portfolio; (iii) map and reconnect the scientific diaspora; (iv) measure and redesign the AI exposure of the national workforce; (v) profile the small- and medium-enterprise sector's technology readiness; and (vi) simulate the marginal returns to public research investment. Crucially, each component already exists in working form; the grant validates the system on Kazakhstani data, deploys it on sovereign infrastructure, and stands up the closed loop from measurement to policy. We therefore propose not a program of promised research but the national deployment of a functioning system.

1 Motivation and Vision

Resource-rich economies face a well-documented development trap: the very abundance that finances the state can crowd out the institutions and incentives required to build a diversified, innovation-driven economy. Kazakhstan's national strategy explicitly seeks to escape this trap by cultivating high-value knowledge industries, commercializing university research, digitizing government and industry, and equipping the workforce for an economy in which artificial intelligence is a general-purpose technology. These are the right objectives. What has been missing is a way to act on them with precision.

A government seeking to diversify confronts a series of concrete questions. Which fields of its national research base are genuinely commercially valuable, and which are strong only academically? Which specific inventions, sitting today in university laboratories, should a sovereign fund back? Where are the country's scientists—at home and in the diaspora—and which collaborations would create the most value? Which occupations are most exposed to AI, and how should those jobs be redesigned rather than merely eliminated? Which firms are ready to adopt new technology, and which require support? And, given a finite research budget, where does the next unit of public investment produce the greatest return?

Historically these questions have been answered, if at all, by committee, anecdote, or imported consultancy. We propose instead to answer them *computationally and continuously*, using instruments that read the nation's scientific and economic text at scale and translate it into decisions. The scientific premise is that the commercial and social potential of research, and the AI exposure of work, are *predictable from text* with useful accuracy; the engineering premise is that the instruments to do so already exist and can be pointed at Kazakhstan. This proposal joins the two.

2 Intellectual and Technical Foundations

The proposed system rests on four foundations that the principal investigators have already built and validated.

2.1 A validated science of commercial potential (Masclans et al.)

The intellectual core is a machine-learning approach that predicts the commercial, scientific, and social potential of scientific work directly from its text, learned from historical relationships between published research and downstream outcomes such as patenting, licensing, and firm formation. This body of work establishes that "valuable science" is identifiable at scale and in advance, rather than only in retrospect—the necessary precondition for any national commercialization strategy. It supplies both the measures the system uses and the scholarly validation that those measures are meaningful.

2.2 A deployed research-scoring platform

These measures are operationalized in a production research-assessment platform with an application-programming interface that scores research abstracts along commercial, scientific, and social dimensions and supports a suite of higher-order instruments: institutional benchmarking, invention scoring and ranking, licensing-brief generation, technology-domain scans, collaborator and co-founder identification, and portfolio optimization. The platform is backed by frozen scientific-language-model embeddings with calibrated predictive heads and is served in production; it is, in effect, the measurement layer of the proposed system, already engineered and running.

2.3 A workforce-redesign engine (OpenAI research grant)

Under an OpenAI research grant, the investigators developed instruments that measure the AI exposure of occupations and guide the *redesign* of work— distinguishing what only humans can lead, own, and judge from what AI can search, structure, and translate—and deliver this as a structured, hands-on exercise rather than a static report. This engine converts the abstract anxiety of "automation" into concrete, occupation-level reskilling and job-redesign plans, and is the human-capital counterpart to the science-commercialization layer.

2.4 An authoring and delivery infrastructure

Finally, an authoring and delivery layer provides the means through which these instruments are configured, run with real populations (cohorts of professionals, civil servants, or students), and observed. It already includes multimodal enterprise profiling, live facilitated activities, and—critical for a government partner—the ability to route all computation through an organization's *own* private models and infrastructure (Section 5).

Individually, these are research artifacts and products. Integrated and pointed at a single nation, they become an operating system for the innovation economy. Section 10 demonstrates the measurement layer running live on Kazakhstan's own research base.

3 The National Innovation Operating System

We organize the work into six instruments (work packages, WP1–WP6). Each is built on the foundations above and already exists in working form; the grant's contribution is validation on Kazakhstani data, sovereign deployment, and integration into a closed decision loop. Together they trace the value chain from scientific knowledge to economic activity: *measure* the science, *select* the bets, *connect* the talent, *redesign* the work, *map* the firms, and *allocate* the funds.

WP1 — The Kazakhstan Science-to-Value Atlas

We score the nation's entire research output—every paper, researcher, institution, and field for which text is available through open bibliographic sources—along the commercial, scientific, and social dimensions. The result is a living atlas of where Kazakhstan's commercializable science actually resides: its genuine strengths, its overlooked commercial assets, and its gaps relative to comparator states. This transforms research-policy debate from opinion to evidence and gives ministries a defensible basis for concentrating support.

WP2 — The Sovereign Deep-Technology Portfolio

Applying invention scoring, ranking, and portfolio optimization, we assemble from the national research base a *diversified, fundable* portfolio of high-potential inventions, each accompanied by an automatically generated licensing brief situating it against comparable science and the surrounding patent landscape. The deliverable is a decision-ready shortlist for a sovereign technology fund or a national tech-transfer office—exactly the artifact such bodies currently lack.

WP3 — The Diaspora and Collaboration Engine

We map Kazakhstani-origin scientists worldwide and identify the collaborations and co-founder pairings that would create the most value, given the national research profile. This directly addresses brain drain: rather than lamenting the diaspora, the system provides a concrete plan to reconnect it and to seed science-based ventures.

WP4 — National AI Exposure and Job-Redesign Program

Beginning with the civil service and two priority sectors, we measure the AI exposure of occupations and deliver, at scale, structured job-redesign and reskilling plans. The output is a workforce-modernization blueprint grounded in the specific tasks Kazakhstani workers actually perform, positioning AI as a tool for augmenting the labor force rather than displacing it.

WP5 — The Enterprise Innovation Census

Using multimodal profiling, we build a living panel of the small- and medium-enterprise sector's technology readiness and needs. The panel supports targeting of support programs and, because it is longitudinal, the measurement of technology diffusion over time—an evaluation capability most innovation programs never obtain.

WP6 — The Research-Allocation Simulator

Finally, using value-to-go estimation, we simulate the marginal commercial and social return of public research investment across fields, giving policymakers a principled answer to the question of where the next unit of funding should go, and what reachable ceiling it can be expected to approach. This work package is the system's research frontier and its most novel scholarly contribution.

4 Technical Approach

Methodologically, the system is built from text. Scientific and occupational corpora are embedded using scientific-language models; calibrated predictive heads map these embeddings to the commercial, scientific, and social potential measures validated in the foundational work. Higher-order instruments compose these scores: invention ranking orders disclosures by predicted potential; portfolio optimization selects a set that maximizes aggregate potential subject to a diversity constraint, so that the recommended portfolio is not merely a list of top-scoring items but a genuinely spread set of bets; and value-to-go estimation, trained through self-play over sequences of research and commercialization decisions, estimates not just an item's current value but the value reachable from it—distinguishing already-mature work from work with headroom. The workforce instruments apply the same text-first logic to occupational descriptions and to the artifacts workers produce.

Two properties make the approach suitable for national deployment. First, it is *automatic and continuous*: once pointed at a corpus, it scores it without manual review, and re-scores as the corpus grows. Second, it is *interpretable at the decision level*: every recommendation is accompanied by the evidence—the comparable science, the patent landscape, the specific tasks—that a human decision-maker needs in order to act and to be accountable.

5 Data Sovereignty and Governance

For a government partner, data sovereignty is not a feature but a precondition. The delivery infrastructure supports per-organization configuration of the underlying AI provider: Kazakhstan can route all computation through its own private or self-hosted models, running on national infrastructure, so that sensitive research, enterprise, and workforce data never leave the country or reach a third-party model. The system defaults to this sovereign configuration where the partner requires it and falls back to shared infrastructure only for non-sensitive tasks and only by explicit choice. All personal and enterprise-level data are governed under an agreed data-protection framework, processed for the stated public-policy purposes alone, and retained under the partner's control. This architecture allows the system to deliver national-scale intelligence while keeping the nation's data within the nation's authority.

6 Work Plan and Timeline

The work proceeds in three phases over [24] months. Because the instruments already exist, the early phase emphasizes data integration and validation rather than construction, allowing decision-ready deliverables within the first two quarters.

Phase	**Period**	**Milestones**
Phase I: Foundation	Months 1–6	Data-sharing and sovereignty agreements; ingestion of the national research corpus and institution list; sovereign-model deployment; first Science-to-Value Atlas (WP1) and initial deep-technology portfolio (WP2).
Phase II: Deployment	Months 7–16	Diaspora and collaboration engine (WP3); AI-exposure measurement and first job-redesign cohorts in the civil service and two priority sectors (WP4); enterprise census pilot (WP5).
Phase III: Integration	Months 17–[24]	Research-allocation simulator (WP6); integration of the six instruments into a single decision loop; validation study, policy handbook, and capacity transfer to the partner institution.

7 Expected Outcomes and Policy Impact

The tangible outputs are decision instruments in daily use by the partner ministries: an atlas the research-policy body consults, a portfolio the sovereign fund draws from, a diaspora plan the talent agency executes, a reskilling blueprint the civil-service and labor bodies implement, an enterprise panel the SME-support programs target, and an allocation simulator the science-funding body consults. The intellectual outputs include peer-reviewed validation of the potential and value-to-go measures on a national corpus, and—through WP6—a novel contribution to the economics of public research allocation. The strategic outcome is a repeatable, sovereign capability: an operating system Kazakhstan owns and can run indefinitely, converting its scientific and human capital into economic value long after the grant concludes.

Institution	**City**	**High-CP**	**Mean CP**
Nazarbayev University	Astana	5,094	99
Al-Farabi Kazakh National University	Almaty	3,711	97
L. N. Gumilyov Eurasian National University	Astana	2,025	93
Satbayev University	Almaty	1,068	91
Kazakh-British Technical University	Almaty	621	82
Kazakh National Medical University	Almaty	492	86
M.Auezov South Kazakhstan State University	Shymkent	367	69

Table 1: Kazakhstani institutions ranked by volume of high-commercial-potential research. **High-CP**: count of Kazakhstan-affiliated publications scoring ≥ 0.6 for commercial potential. **Mean CP**: mean commercial-potential score (0–100) of each institution's 100 top-scored outputs.

8 Team and Feasibility

The principal investigator has built the scientific measures, the scoring platform, the workforce-redesign engine, and the delivery infrastructure on which this proposal depends; the co-principal investigator provides the governmental mandate, data access, and institutional pathway to adoption within the Republic of Kazakhstan. Feasibility is unusually high for a proposal of this ambition precisely because its central risk—whether the instruments can be built—has already been retired. What remains is validation on national data and deployment, which the team is positioned to execute immediately.

9 Budget Summary

We request [amount] over [24] months. Principal categories are: personnel for data engineering, validation, and deployment; sovereign compute and model-serving infrastructure; data acquisition and integration; fieldwork for the enterprise census and workforce cohorts; and dissemination and capacity transfer to the partner institution. A detailed budget and justification are provided in the supplementary materials.

10 A Working Demonstration: Kazakhstan's Commercial Research Landscape

To show that the system's measurement layer is a functioning instrument rather than a promise, we applied it to Kazakhstan's research base: every Kazakhstan-affiliated publication and researcher was scored for commercial, scientific, and social potential. The results below—a national map, an institutional ranking, the specific topical fronts on which each institution is commercially strong, and the leading researchers behind that work—were produced automatically and are continuously refreshable as the corpus grows. They correspond to Work Packages 1 and 2.

Figure 1 locates the nation's commercially promising science; Table 1 ranks the institutions that produce it. Crucially, Table 2 moves past broad disciplines to the *specific* research fronts—keyword signatures—on which each institution is strong, the resolution at which commercialisation decisions are actually made.

Finally, a commercialisation strategy is executed by people. Table 3 identifies the leading Kazakhstan-based researchers by the commercial potential of their work, with the topics they work on—the human capital the strategy must engage, retain, and connect to industry and to the diaspora.

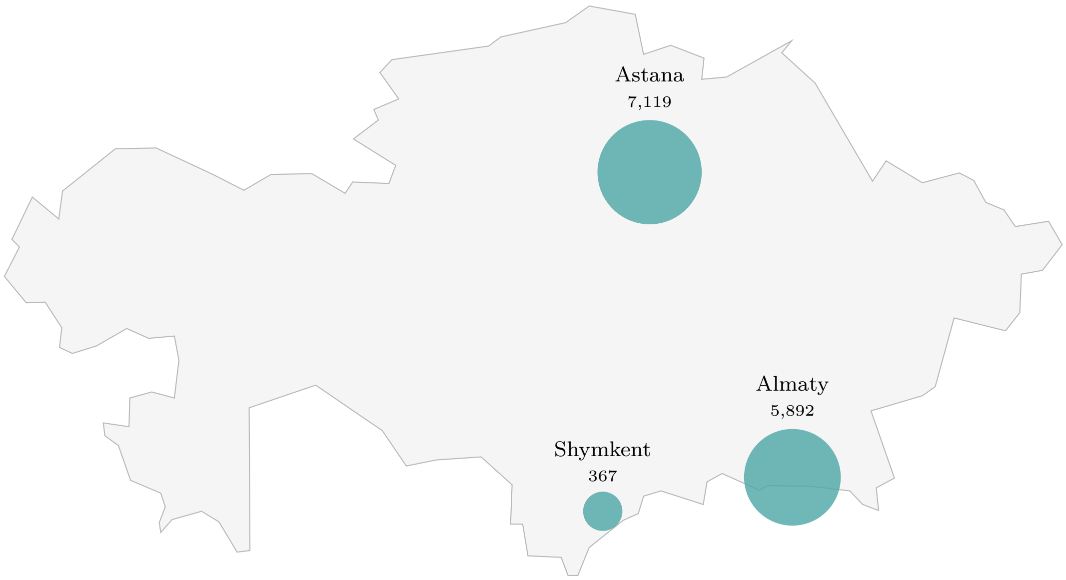

Figure 1: Geography of high-commercial-potential research across Kazakhstan. Bubble area is proportional to the number of Kazakhstan-affiliated publications with commercial potential ≥ 0.6 at each city's institutions. Output concentrates sharply in Astana and Almaty, with Shymkent emerging—a spatial concentration that is itself a finding for regional innovation and talent policy.

References

[1] Masclans, [Initials], Hasan, S., et al. *[Title: machine-learning identification of commercially valuable science].* Working paper, [Year].

[2] Hasan, S. et al. *Redesigning work in the age of AI: measuring occupational exposure and guiding job redesign.* OpenAI research grant, [Year].

Institution	Specific topical strengths (most frequent keywords among its highest commercial-potential work)
Nazarbayev University	Fiber Optic Sensors, Distributed Sensing, Biosensors, Haptic Interfaces, Audio-Visual Speech Recognition, Cancer Immunotherapy, Car T Cells
Al-Farabi Kazakh National University	Neural Machine Translation, Two-Dimensional Materials, Machine Translation, Multilingual Neural Machine Translation, Language Modeling, Self-Assembled Monolayers, Translation
L. N. Gumilyov Eurasian National University	Membrane Fouling, Nanocrystal Synthesis, Wastewater Treatment, Immunity, Deep Learning, Sustainable Water Treatment, Membrane Technology
Satbayev University	Polymeric Gels, Controlled Polymerization, Adsorption, Hydrogels, Hydrogen Production, Graphene, Surfactants
Kazakh-British Technical University	Named Entity Recognition, Information Retrieval, Color Psychology, Aesthetic Intelligence, Ionic Liquids, Detection, Language Modeling
Kazakh National Medical University	Asthma, Aging, Mesenchymal Stem Cells, Nk Cell Therapy, Nk Cell Development, Car T Cells, Nk Cell Recognition
M.Auezov South Kazakhstan State University	Nanoencapsulation, Dyeing, Integro-Differential Equations, Plant Growth, Wastewater Treatment, Inverse Problems, Inverse Scattering Theory

Table 2: The concrete research fronts, not broad disciplines, where each institution's commercially promising work concentrates. These keyword signatures reveal genuine specialisation: photonic and fiber-optic sensing and cell therapy at Nazarbayev; batteries, graphene, and hydrogen at Satbayev; machine translation and two-dimensional materials at Al-Farabi; membranes and water treatment at Gumilyov; stem-cell and NK/CAR-T therapy at the medical university.

Researcher	Institution	Topics	Pubs	CP
Vesselin N. Paunov	Nazarbayev University	Pickering Emulsions, Microemulsions, Nanoparticles, Droplet Formation	137	90
Daniele Tosi	Nazarbayev University	Fiber Optic Sensors, Refractive Index Sensor, Distributed Sensing, Long-Period Gratings	96	88
Yuliya Semenova	Nazarbayev University	Fiber Optic Sensors, Refractive Index Sensor, Sensor Technology, Whispering Gallery Mode	120	88
Zhumabay Bakenov	Nazarbayev University	Battery Materials, Nanostructured Cathodes, Nanostructured Anodes, Lithium-Sulfur Batteries	122	88
Woojin Lee	Nazarbayev University	Reductive Dechlorination, Nano Zero-Valent Iron, Metastatic Pancreatic Cancer, High-Efficiency Solar Cells	546	86
Kyunghwan Oh	Nazarbayev University	Microstructured Optical Fibers, Fiber Lasers, Nonlinear Optics, Fiber Optic Sensors	100	86
Thierry Djenizian	Al-Farabi Kazakh National University	Anode Materials, Nanostructured Anodes, Electrode Materials, Battery Materials	87	85
Michael Hippler	Institute of Plant Biology and Biotechnology	Photosynthetic Acclimation, Photosystem II, Light-Harvesting, Photocycle Dynamics	98	84
Ferdinand Molnár	Nazarbayev University	Vitamin D, Epigenetic Reprogramming, Tumor Evolution, Cancer Epigenetics	68	84
Danil Bukhvalov	Satbayev University	Graphene, Metal-Organic Frameworks, Single-Molecule Magnets, Graphene Oxide	197	84

Table 3: The people behind the science: leading Kazakhstan-based researchers by the commercial potential of their work (of 2,369 indexed Kazakhstan-affiliated researchers with sustained output). **CP**: mean commercial potential (0–100). This is the human capital a commercialisation strategy must engage, retain, and connect.